Working with powers — Simplifying · Index laws · Expanding · Factorising

No calculator. ★ = challenge question.

A · SIMPLIFYING AND INDEX LAWS

1 Simplify

a $9y^2 - 4y^2$ **b** $3x + 2x^3 + 5x - x^3$ **c** $5d^3 - 3 + 2d^3 + 7 - d^2$

2 Write each expression as a single power.

a $x^5 \times x^4$ **b** $\dfrac{y^{12}}{y^4}$ **c** $(z^3)^4$ **d** $m^2 \times m^6 \times m$ **e** $\dfrac{a^5 \times a^3}{a^6}$

3 Simplify

a $4a \times 6b$ **b** $3n \times (-2n)$ **c** $5p^2 \times 2p^3$ **d** $\dfrac{20x^6}{4x^2}$ **e** $(2y)^3$

B · EXPANDING BRACKETS

4 Expand

a $p(p^2 + 3p)$ **b** $3m^2(m - 5)$ **c** $2x(4x^2 - x + 6)$

5 Expand and simplify

a $4(x + 2) + 3(x - 1)$

b $5(2y - 3) - 2(y + 4)$

c $x(x + 4) + 2x(3x - 1)$

6 Show that $2x(3x + 1) + x(x - 2) \equiv 7x^2$.

C · FACTORISING

7 Factorise

a $3x + 15$ **b** $10y - 25$ **c** $18 - 12a$

8 Factorise completely

a $6y^2 + 9y$ **b** $8x^3 - 12x$ **c** $10w^3 + 15w^2 - 5w$

9 Sam factorises $12x^2 + 18x$ and writes $6(2x^2 + 3x)$. Explain why this is not factorised *completely*, and write the complete factorisation.

D · SUBSTITUTING AND SOLVING

10 Decide whether each statement is an **identity** (true for every value) or an **equation** (true for particular values). If it is an equation, solve it.

a $2x + 3x + 1 = 5x + 1$

b $4x - 2 = 10$

c $3(y + 2) = 3y + 6$

11 Find the value of each expression when $x = 5$, $y = -3$ and $z = 2$.

a $3(x - y) + 2z$ **b** $2x^2 - yz$ **c** $(y + z)^2 - x$

12 Solve $4(x - 1) = 50 - 2(x + 3)$

13 A rectangle 4 cm by $(3x - 7)$ cm has the same area as a rectangle 2 cm by $(5x + 2)$ cm. Work out x and the area.

14 The formula $v^2 - u^2 = 2as$ links the starting speed u, the final speed v, the acceleration a and the distance s of a moving object. A car starts from rest ($u = 0$) and accelerates at 1.5 m/s² until it reaches 12 m/s. How far has it travelled?

Check: like terms need the same power · index laws need the same base · a bracket times a term — every term inside · "completely" means the HCF, letters included.